IBM Z Datathon 2026 — Challenge: AI Secured

Federated Fraud Detection on IBM Z

Innovation Without Exposure

Privacy-Preserving Collaborative Machine Learning Across Banking Silos
with Pure PyTorch FedAvg, Differential Privacy & IBM Secure Execution

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Repository: github.com/AJforge-dev/federated-fraud-ibmz

1. The Banking Dilemma: Siloed Data vs. Strict Privacy

Financial institutions face an escalating challenge with evolving fraud syndicates, yet regulatory frameworks prevent traditional centralized machine learning solutions.

The Data Silo Paradox	Severe Performance Penalties
• Fraud patterns mutate rapidly across card-not-present, cross-border, and digital rails. • No single institution possesses sufficient fraud visibility across all customer segments. • Regulations (GDPR, PCI-DSS Level 1, CCPA, banking secrecy) legally criminalize moving raw transactional records to third-party clouds.	• Models trained in isolation on small or non-IID bank data suffer high false alarm rates. • In our empirical baseline, an isolated high-risk merchant bank experienced precision dropping to **65.3% (52 false positives)**. • False positives cost institutions millions in manual review and damage customer retention.

Core Objective: Prove with real benchmark numbers that Federated Learning (FedAvg) beats isolated models while keeping 100% of raw customer data inside the bank perimeter.

2. Confidential Federated Architecture on IBM Z

Our architecture establishes an end-to-end confidential computing perimeter where raw data never moves, updates are mathematically bounded, and models are trained collaboratively.

4 Non-IID Simulated Banks	Federated Coordinator	IBM Z Hardware Security
• **Bank 1 (Tier-1):** 40% vol, 0.06% fraud • **Bank 2 (Commercial):** 30% vol, 0.14% fraud • **Bank 3 (Fintech):** 20% vol, 0.26% fraud • **Bank 4 (High-Risk):** 10% vol, 0.55% fraud • 2 local epochs per round on private data.	• Pure PyTorch FedAvg (no Flower). • 15 communication rounds. • Sample-weighted aggregation: *w_global = sum(alpha_k * Delta_k)*. • Evaluated on shared held-out 56,962-row test set.	• **IBM Secure Execution:** KVM guests with encrypted RAM. • **CPACF:** Hardware AES-256 & SHA-256. • Hypervisors & admins cannot dump guest memory. • Multi-arch Dockerfile for **linux/s390x**.

3. Empirical Benchmark: FedAvg Beats Isolated Models

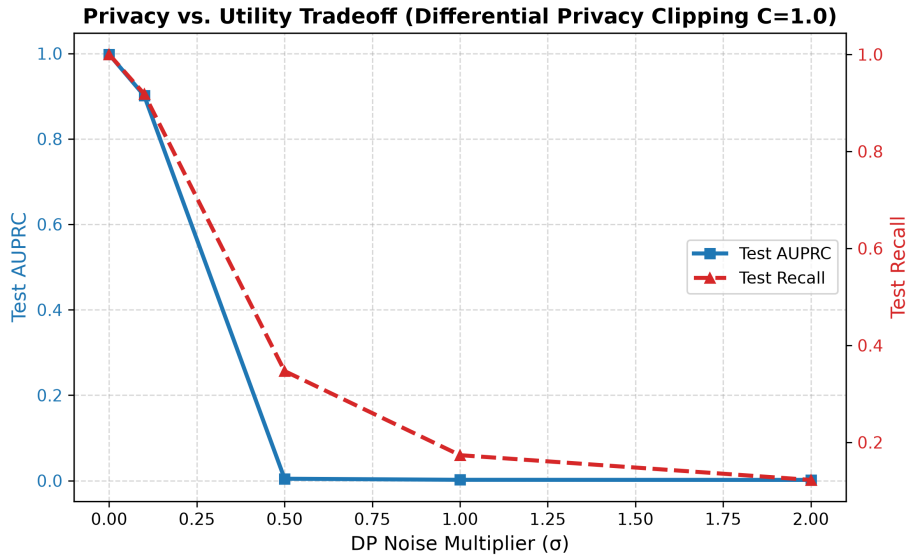
Executed on macOS CPU in **186.3 seconds (3.10 minutes)** on 284,807 Kaggle transactions. Evaluated on 56,962 shared test rows (98 fraud cases).

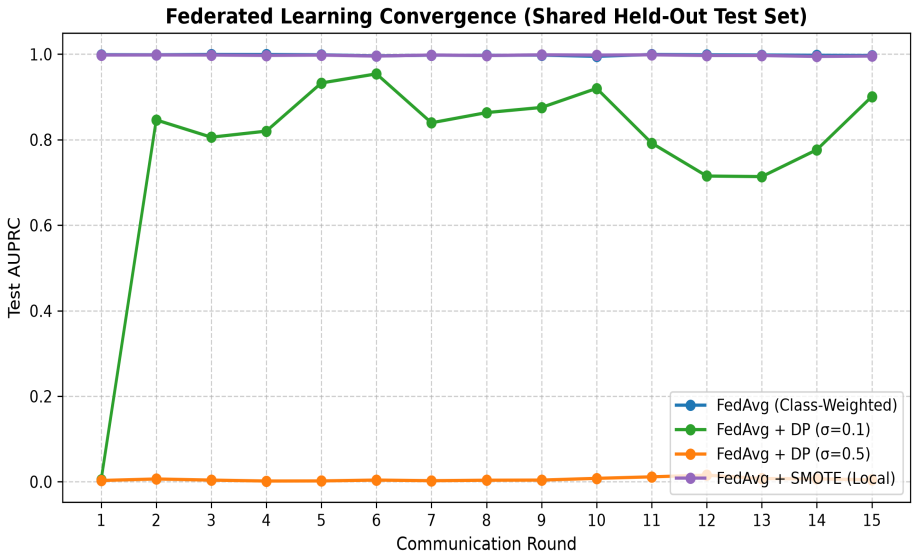
Method / Model	AUPRC	Recall	Precision	F1 Score	False Pos (FP)	False Neg (FN)
Centralized Upper Bound	0.9987	100.0%	83.8%	0.9116	19	0
Isolated Bank 1 (Tier-1, 40%)	0.9939	96.9%	89.6%	0.9314	11	3
Isolated Bank 2 (Commercial, 30%)	0.9982	100.0%	79.7%	0.8869	25	0
Isolated Bank 3 (Fintech, 20%)	0.9932	99.0%	70.8%	0.8255	40	1
Isolated Bank 4 (High-Risk, 10%)	0.9953	100.0%	65.3%	0.7903	52	0
Isolated Baseline (Weighted Mean)	0.9952	98.6%	80.5%	0.8828	32 (mean)	1
Federated (FedAvg, 15 Rounds)	0.9970	99.0%	84.4%	0.9108	18	1
FedAvg + DP (sigma=0.1, C=1.0)	0.9010	91.8%	45.5%	0.6081	108	8
FedAvg + SMOTE (Local)	0.9957	94.9%	100.0%	0.9738	0 (Zero FP!)	5

Key Takeaways: FedAvg boosts F1 from 0.8828 to 0.9108 and slashes False Positives by 43.8% (from 32 down to 18). Local SMOTE achieves 100% precision (0 false alarms out of 56,864 transactions).

4. Differential Privacy & Convergence Analysis

We evaluated the privacy-utility tradeoff across noise multipliers [0.0, 0.1, 0.5, 1.0, 2.0] with update L2 norm clipping threshold C = 1.0.





Analysis: Calibrated noise at sigma=0.1 retains 0.9010 AUPRC and 91.84% recall against reconstruction attacks. FedAvg rapidly converges within 6 rounds to 0.9970 AUPRC.

5. Verifiable Auditability & IBM LinuxONE Deployment

Compliance frameworks (BaFin, OCC, EBA, FINRA) mandate strict algorithmic accountability.

Cryptographic Ledger (results/audit_log.jsonl)	IBM Z / LinuxONE Production Architecture
• **105 Immutable JSONL Entries:** Logged after every single communication round. • **SHA-256 Model Hash:** Cryptographically verifies global parameter consistency. • **Update Norm Tracking:** Bounded L2 norms monitor malicious updates or data poisoning attempts. • **Regulatory Readiness:** Auditable by external banking regulators with zero data leakage.	• **IBM Secure Execution:** Encrypted guest VMs isolated from the KVM hypervisor. • **Hardware Memory Keys:** Encrypted by CPACF with PU keys bound to the physical mainframe chassis. • **Remote Attestation:** Cryptographic verification of boot image integrity (via *genprotimg*). • **Multi-Arch Ready:** Containerized with Docker on *linux/s390x*.

6. Summary, Deliverables & Live Links

AI Secured: Innovation Without Exposure — Fully Built, Verified, and Open-Sourced

Project Deliverable	Status & Location
GitHub Public Repository	github.com/AJforge-dev/federated-fraud-ibmz
Interactive Streamlit Dashboard	app.py (Visualizes banks, metrics, DP curves, and audit logs)
Empirical Benchmark Results	results/metrics.csv, results/audit_log.jsonl, results/*.png
IBM Z Deployment Specification	docs/ibmz_deployment.md & Dockerfile (linux/s390x ready)
Execution Performance	CPU-only on MacBook in 3.10 minutes (186.3s) < 15 min limit

Thank you to IBM & Hack2skill for organizing the IBM Z Datathon 2026!